



Data Glacier

Your Deep Learning Partner

Exploratory Data Analysis

Project: G2M Insight For Cab Investment Firm

Data Glacier Virtual Internship

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Agenda

- Problem Statement
- Dataset Information
- Exploratory Data Analysis(EDA)
- Hypothesis Test
- Conclusion

Problem Statement

- XYZ is a private equity firm in the US. Due to remarkable growth in the Cab Industry in the last few years and multiple key players in the market, it is planning for an investment in the Cab industry.
- **Objective:** Provide useful information to assist XYZ organization in choosing which business to invest in.

Dataset Information

- **Cab_Data.csv:** This file includes details of transactions for 2 cab companies.
- **Customer_ID.csv:** This is a mapping table that contains a unique identifier that links the customer's demographic details.
- **Transaction_ID.csv:** This is a mapping table that contains transaction to customer mapping and payment mode.
- **City.csv:** This file contains a list of US cities, their population, and the number of cab users.
- **holidays.csv:** This file contains the holiday list.

Dataset Information (Data Visualization)



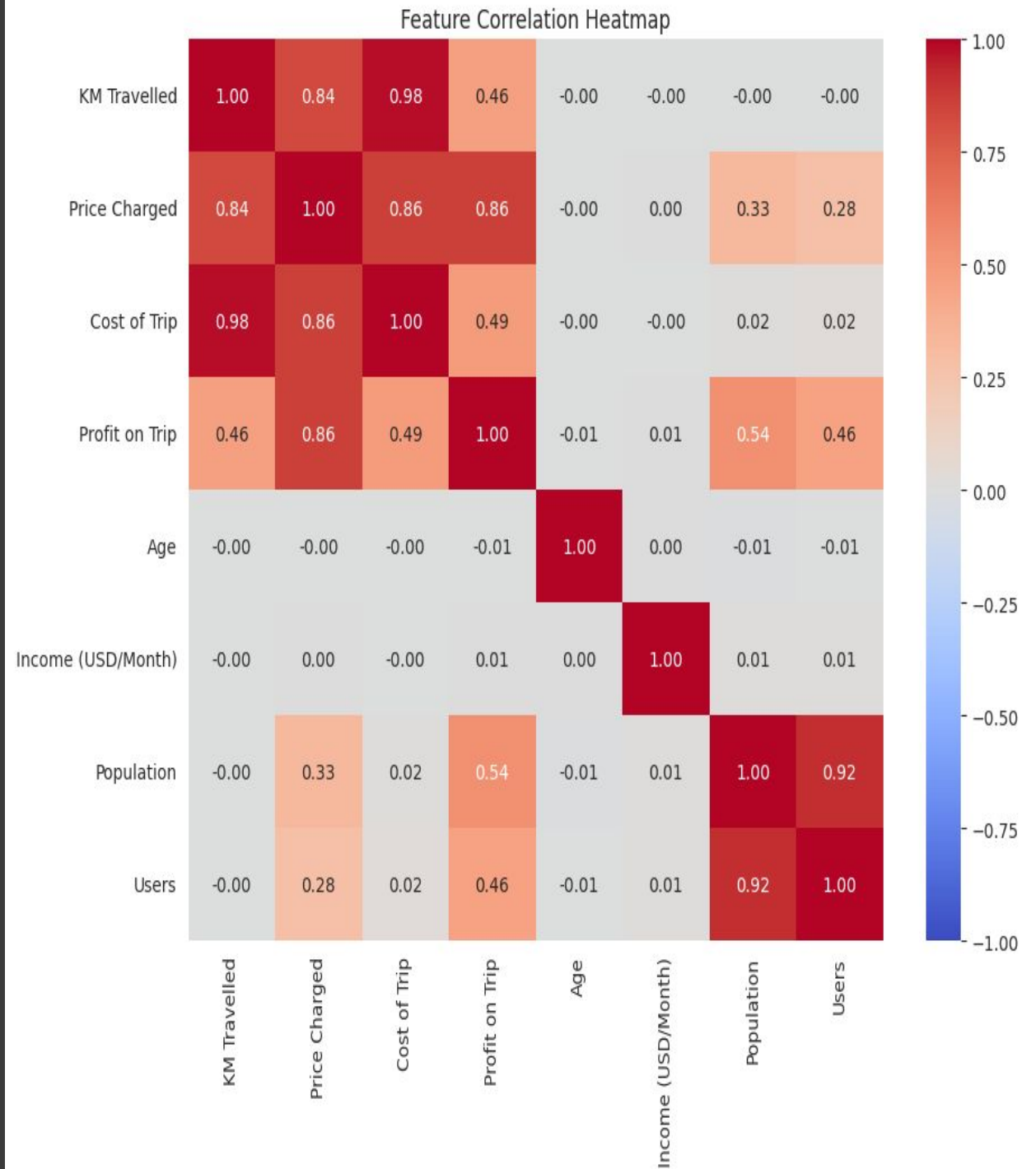
Dataset Information (Data Correlation)

The Strongest relationships are:

- KM Travelled ↔ Cost of Trip (0.98)
- Population ↔ Users (0.92)
- Price Charged ↔ Profit on Trip (0.86)

Top 5 Features Correlated with Profit on trip:

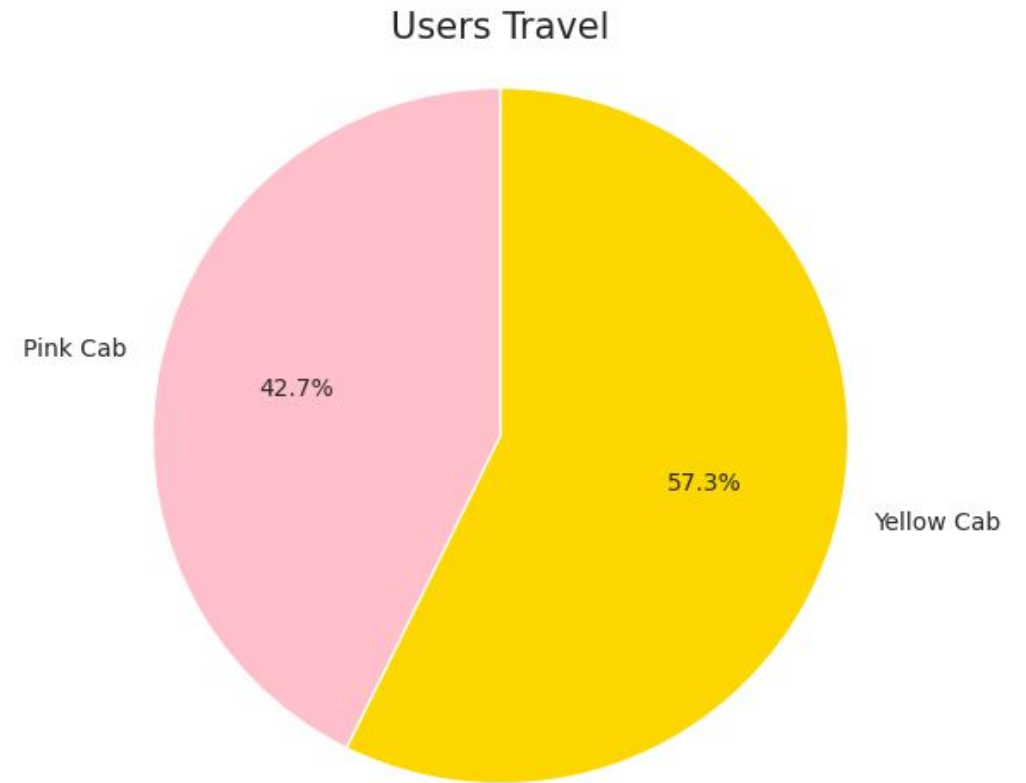
- Price Charged 0.864154
- Population 0.544079
- Cost of Trip 0.486056
- KM Travelled 0.462768
- Users 0.457758



EDA

Which Company has more Users?

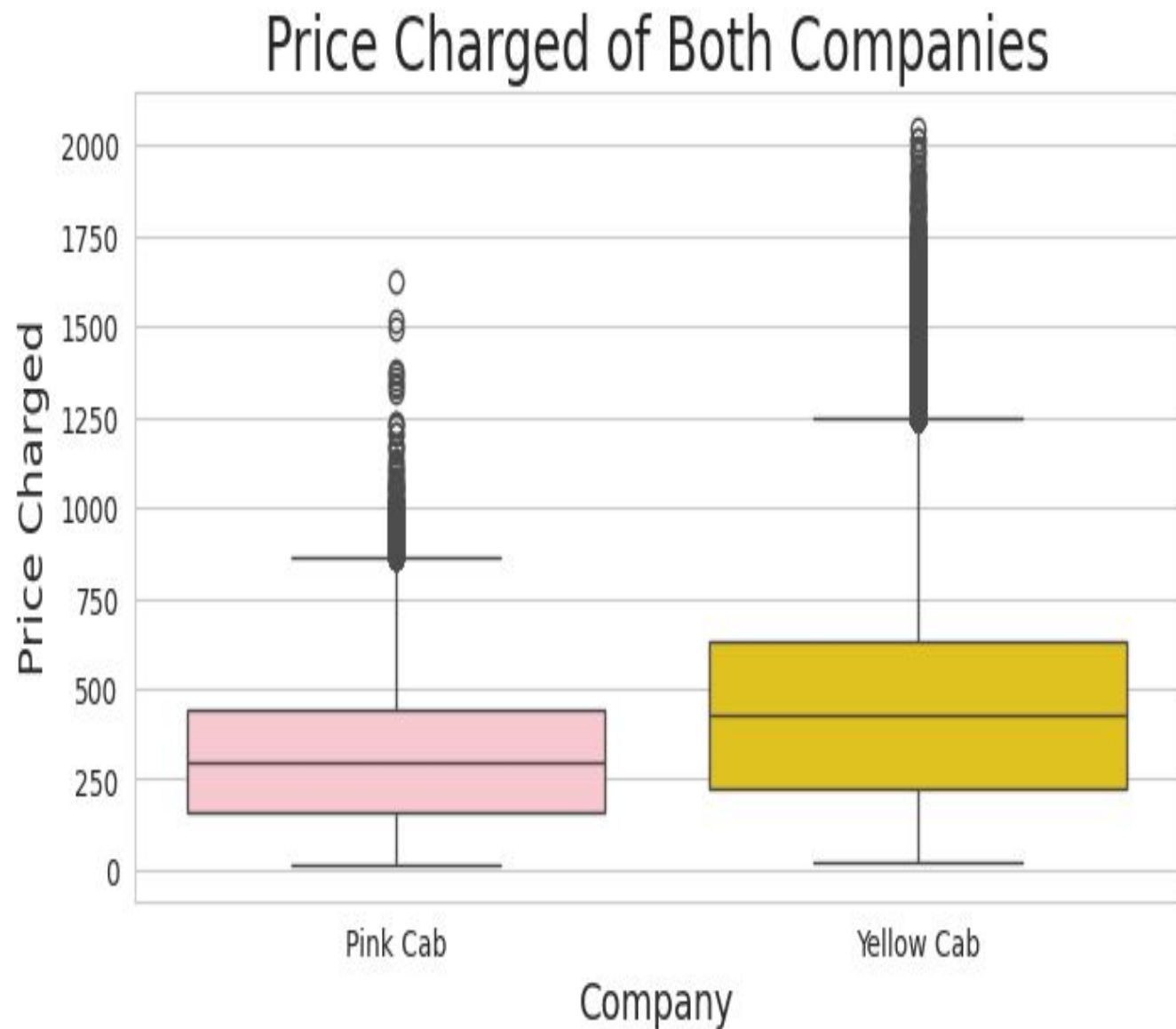
- As we can see, people prefer to take Yellow Cab rides over Pink Cab rides.



EDA

Which Company has a high price charged?

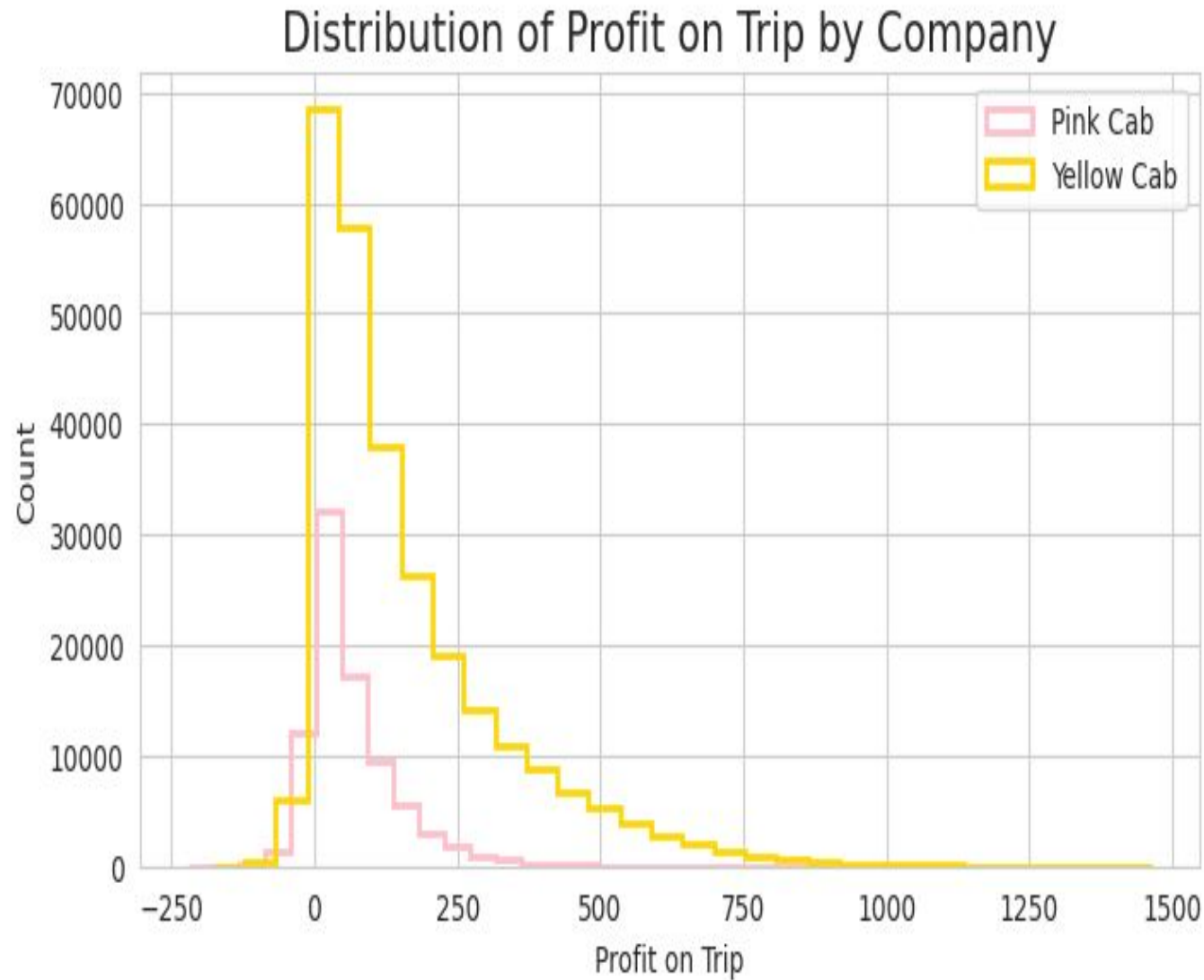
- As we can see, Yellow Cab charges a higher price than Pink Cab.



EDA

Which Company has a high profit on trip?

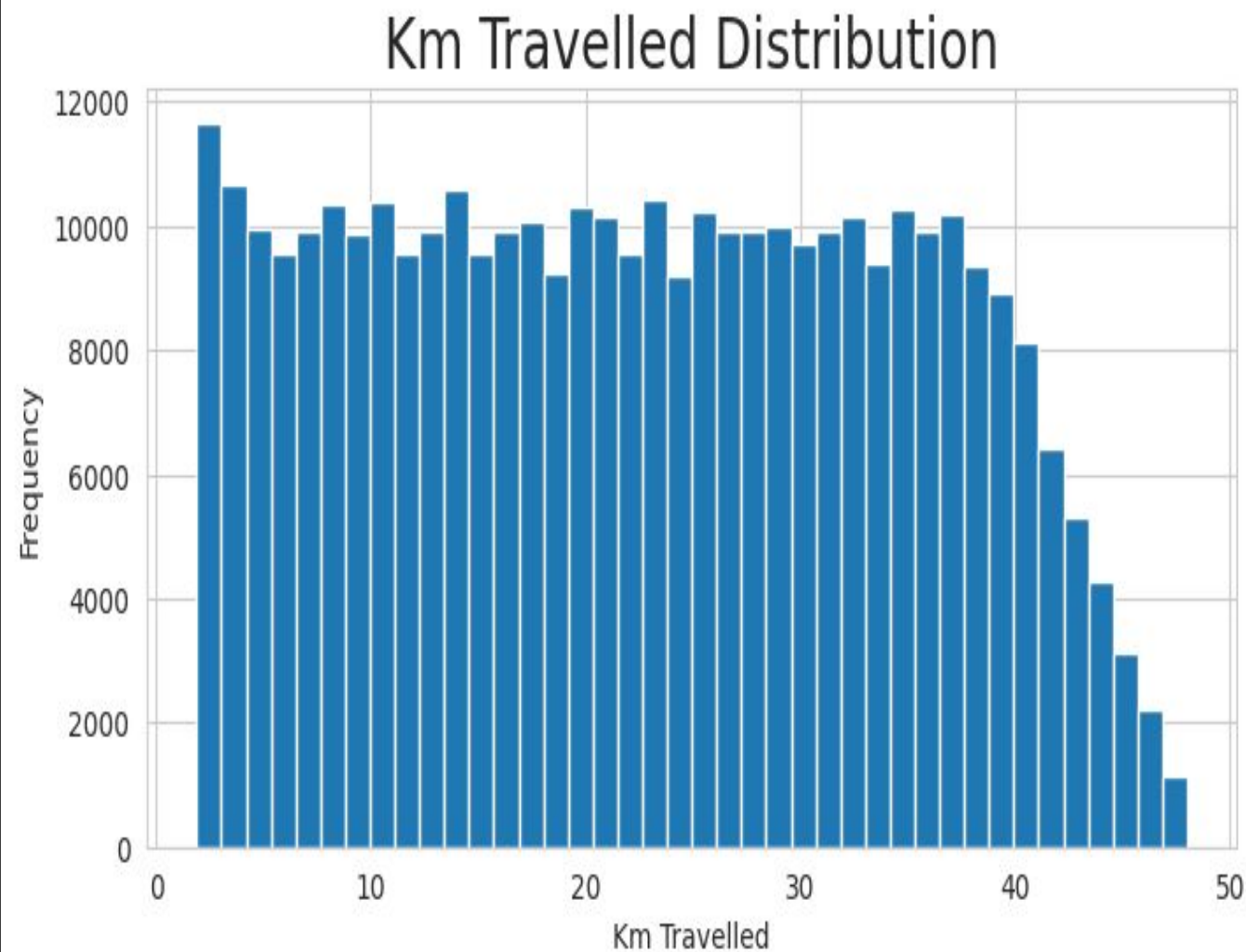
- Profit Distribution accounts larger for Yellow Cab.



EDA

KM Travelled Distribution

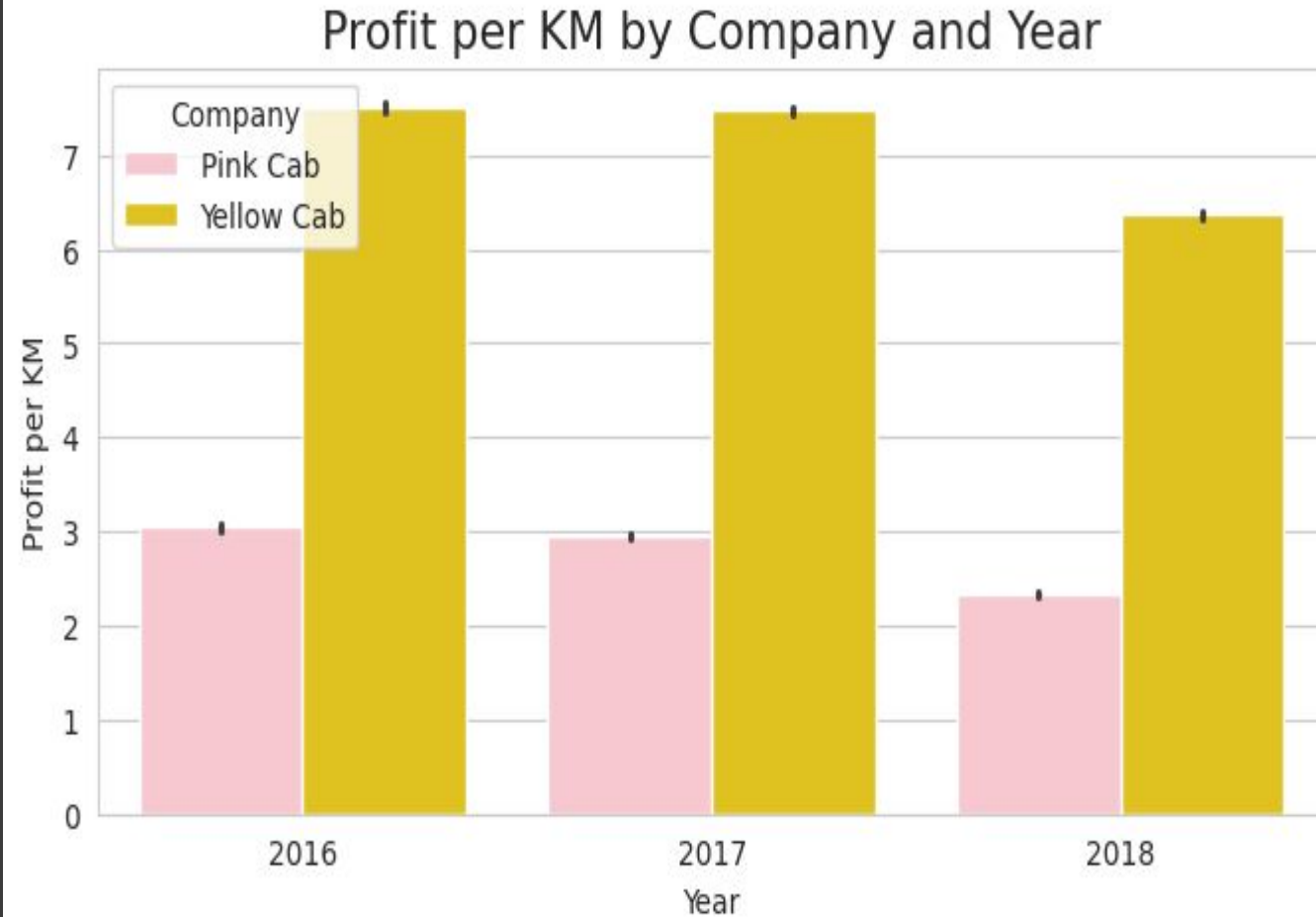
- Most of the rides are between 2 to 48 KM



EDA

Which company has higher profit per KM by year?

- Profit per KM values are higher in 3 years for Yellow cab than pink cab



EDA

Which company has higher profit per KM in holiday?

- Profit per KM values are higher in holiday or others day for Yellow cab than pink cab.

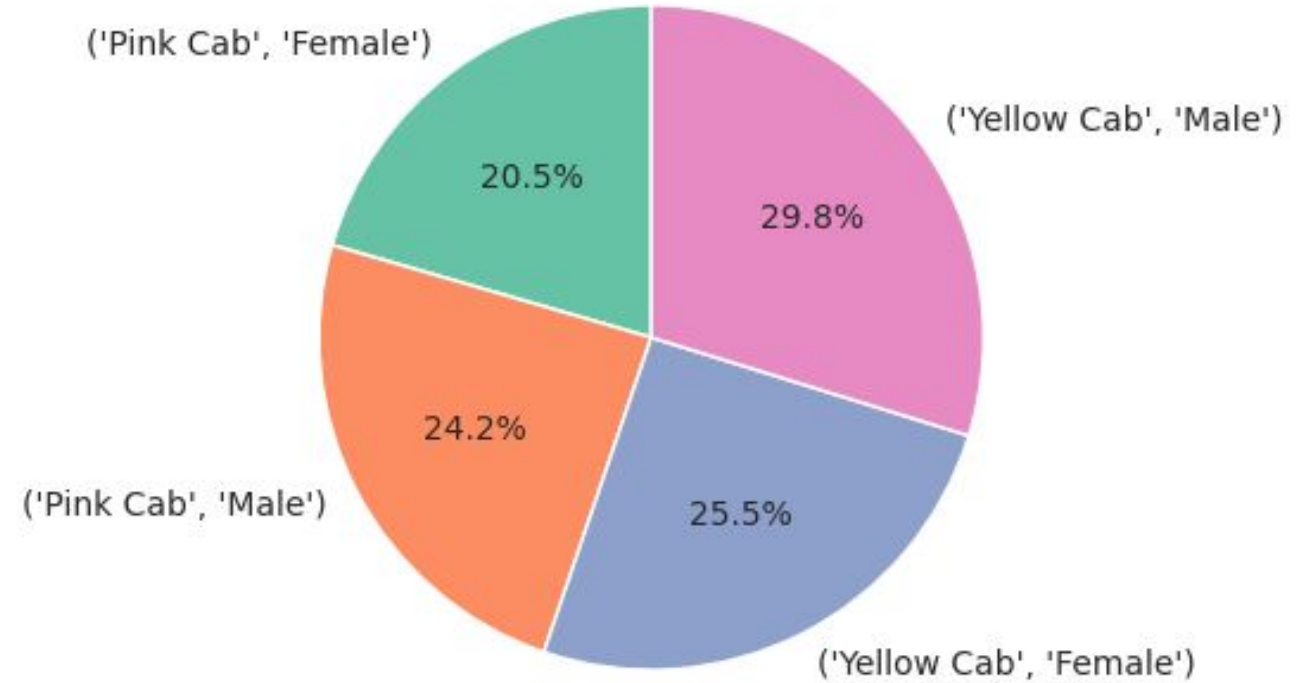


EDA

Users w.r.t Gender

- Male users prefer more to travel in Cab

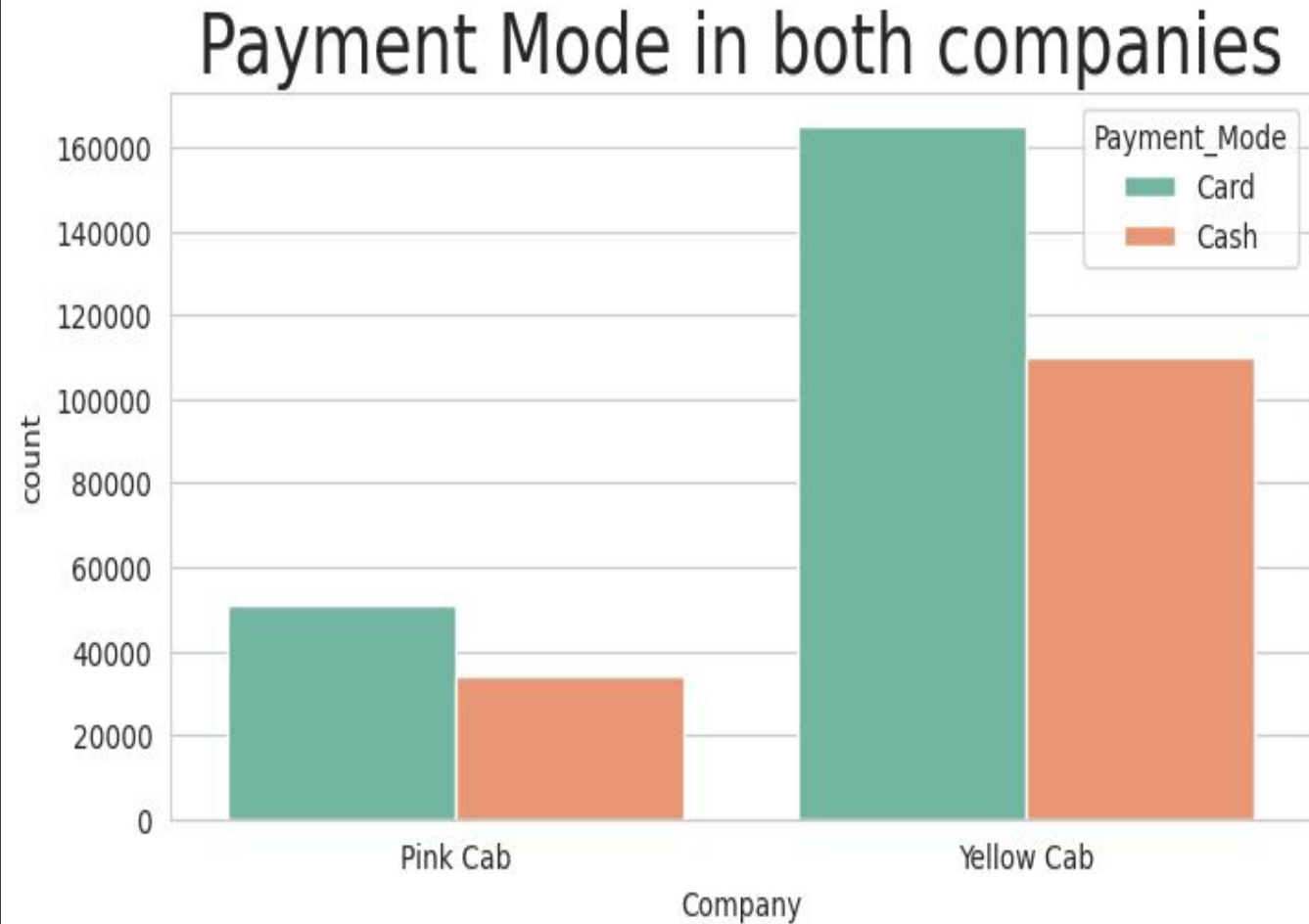
Customer share per gender per cab



EDA

Mode of Payment

- It is evident that users would rather use a card to make payments than cash.



EDA

Profit Margin w.r.t Time

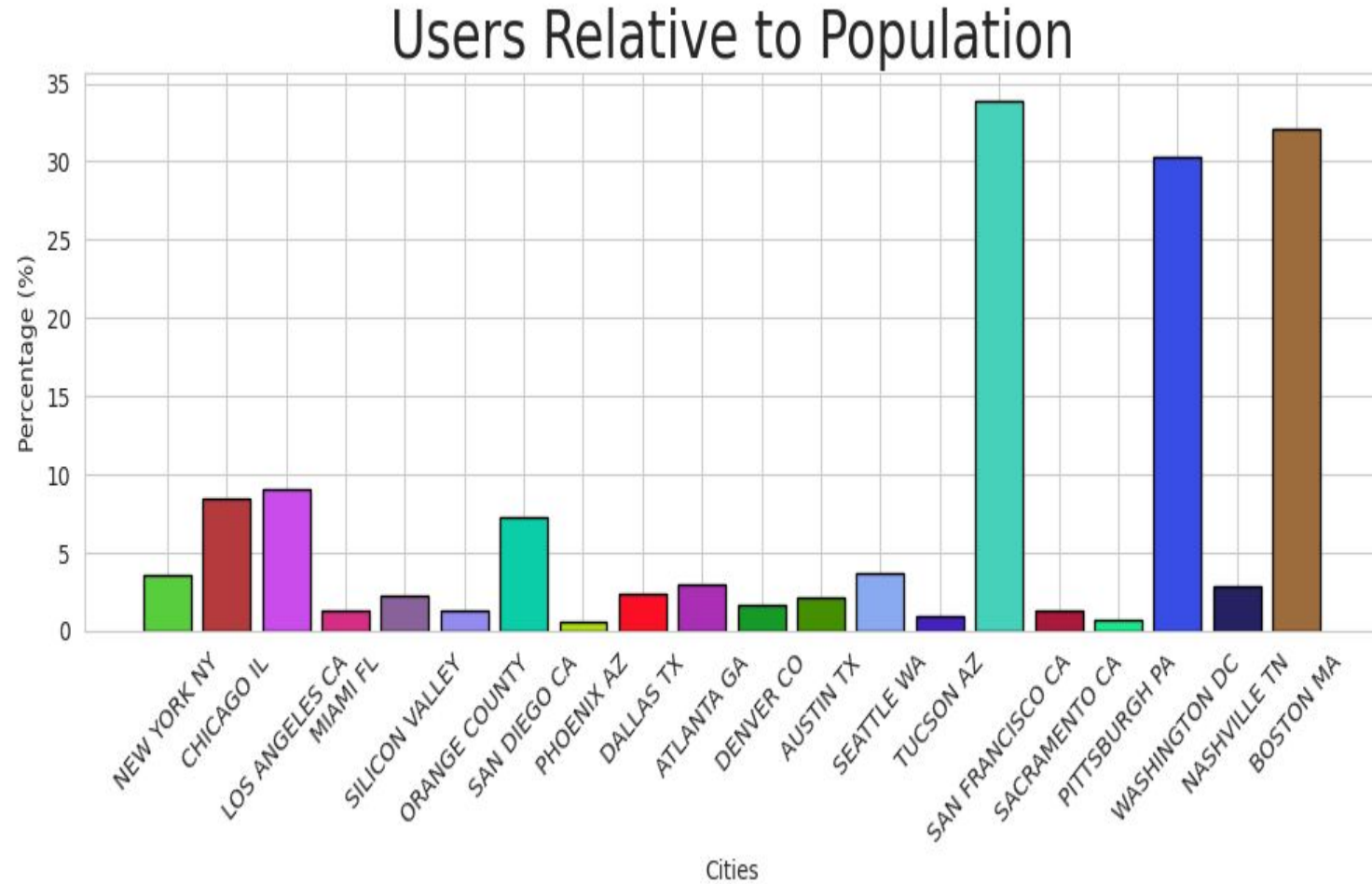
- The profit margin decreased w r t year
- The profit margin varies w r t month



EDA

Users w.r.t Population

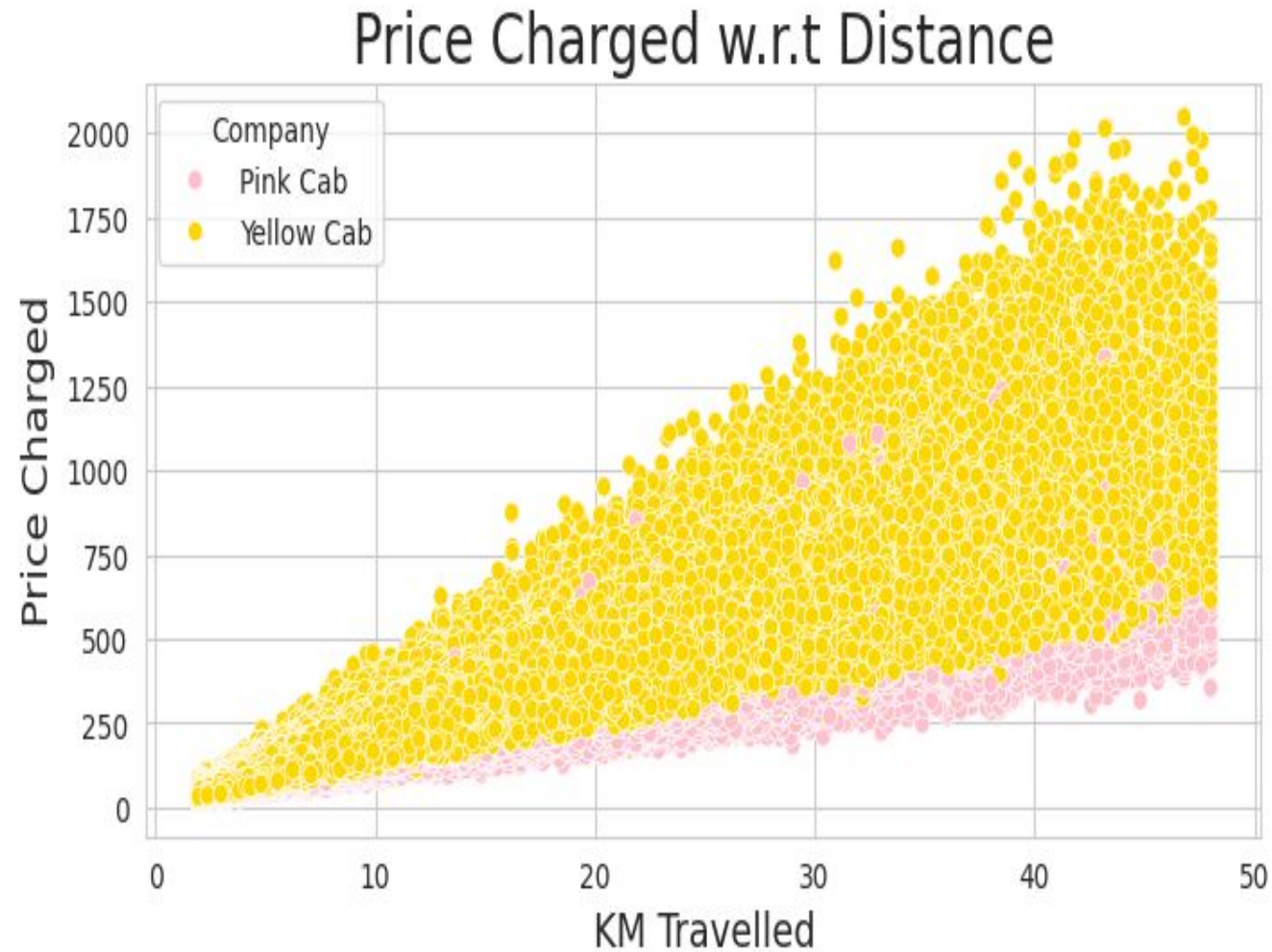
- As we can see in the cities of San Francisco, Washington, and Boston more than 30 of the population use cab service



EDA

Price Charged w.r.t Distance

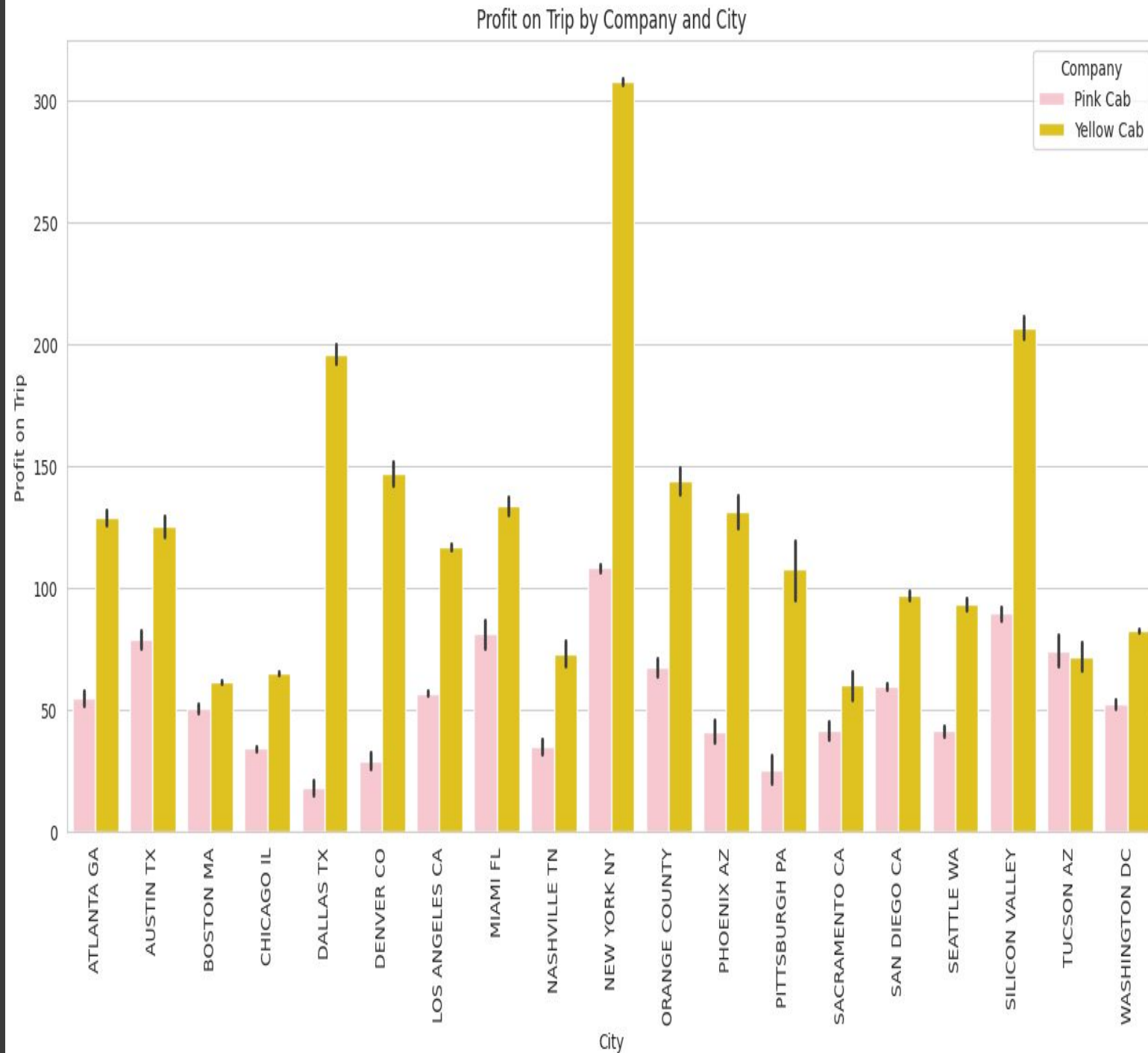
- As expected, we may observe a linear relationship between the price charged and the kilometers traveled. Nevertheless, Yellow Cab charges more than Pink Cab.



EDA

Which Cities have higher profit on trip?

- DALLAS, NEW YORK, SILICON VALLEY



Hypothesis Test

Hypothesis 1: Is there any difference in profit regarding Company

- H_0 : There is no difference in profit.
- H_1 : There is difference in profit.

Test Result:

```
➡ Pink Cab transactions: 84711  
   Yellow Cab transactions: 274681  
   T-statistic: -230.996, P-value: 0.00000  
   Reject  $H_0$ : There is difference in profit ( $H_1$ )
```

Conclusion: There is significant difference in "Profit" between different company.

Hypothesis Test

Hypothesis 2: Is there any difference in KM travelled regarding Company

- H_0 : There is no difference in travel distance.
- H_1 : There is difference in travel distance.

Test Result:

```
→ Pink Cab trips: 84711  
Yellow Cab trips: 274681  
T-statistic: -0.200, P-value: 0.84171  
Fail to reject  $H_0$ : No significant distance difference ( $H_0$ )
```

Conclusion: There is no significant difference in "KM Travelled" between different company.

Hypothesis Test

Hypothesis 3: Is there any difference in profit regarding Gender

- H_0 : There is no difference regarding Gender in both cab companies
- H_1 : There is difference regarding Gender in both cab companies

Test Result:



```
Pink Cab - Male transactions: 47231, Female transactions: 37480
T-statistic: 1.575, P-value: 0.11515
Variance assumption: Equal
Fail to reject  $H_0$ : No significant profit difference by gender for Pink Cab ( $H_0$ )

Yellow Cab - Male transactions: 158681, Female transactions: 116000
T-statistic: 10.375, P-value: 0.00000
Variance assumption: Unequal
Reject  $H_0$ : Significant profit difference by gender for Yellow Cab ( $H_1$ )
Male customers generate higher profit on average.
```

Conclusion: Male customers generate higher profit on average for Yellow cab

Hypothesis Test

Hypothesis 4: Is there any difference in Profit regarding Age

- H_0 : There is no difference regarding Age in both cab companies
- H_1 : There is difference regarding Age in both cab companies

Test Result:



Pink Cab - ≤ 60 transactions: 80125, > 60 transactions: 4586

T-statistic: 1.317, P-value: 0.18796

Variance assumption: Equal

Fail to reject H_0 : No significant profit difference by age for Pink Cab (H_0)

Yellow Cab - ≤ 60 transactions: 260356, > 60 transactions: 14325

T-statistic: 5.267, P-value: 0.00000

Variance assumption: Unequal

Reject H_0 : Significant profit difference by age for Yellow Cab (H_1)

≤ 60 customers generate higher profit on average.

Conclusion: It appears that the Yellow Cab company provides discounts to its clients who are over 60.

Hypothesis Test

Hypothesis 5: Does Profit differ between cash and card payments for each cab company

- H_0 : There is no difference in average profit between cash and card payments.
- H_1 : There is a difference in average profit between cash and card payments.

Test Result:



```
Pink Cab - Cash transactions: 33992, Card transactions: 50719  
T-statistic: -0.266, P-value: 0.79005  
Variance assumption: Equal  
Fail to reject  $H_0$ : No significant profit difference by payment mode for Pink Cab ( $H_0$ )  
  
Yellow Cab - Cash transactions: 109896, Card transactions: 164785  
T-statistic: 1.051, P-value: 0.29331  
Variance assumption: Equal  
Fail to reject  $H_0$ : No significant profit difference by payment mode for Yellow Cab ( $H_0$ )
```

Conclusion: There is no difference in average profit between cash and card payments.



- **Yellow Cab Company is better than Pink Cab Company for investment. Because:**
 - 1) Profit Margin
 - 2) More Users
 - 3) More transactions per Year

Thank You